

The Target Was a Comb Filter

Measurement validity in fitting a modal drum model to a recorded tom

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Abstract. A modal two-headed drum model was fitted to one recorded tom by minimising an eight-term perceptual distance with a swarm optimiser, under each of two excitation models. The fits converged, ranked consistently, and agreed with a listening test on their ordering — and none of them sounded like the recording. Auditing the measurement rather than the search found the reason. The reference is a stereo pair of one hit whose second channel lags the first by 69 samples at 44.1 kHz; the harness reduced it to mono by averaging, which is a comb filter with a notch at 320 Hz and a peak at 639 Hz. The dense cluster of nine partials between 476 and 700 Hz that had been treated as the model’s outstanding deficiency is largely an artefact of that sum: de-combed, the reference has five detected partials with two in the band, against the model’s five. Two further defects compounded it. The modal budget is a fixed mode **count**, so the rendered bandwidth scales with $\sqrt{T}$ and the fitted bank — tuned down to reach the reference’s ring time — has no modes at all above 467 Hz. And the excitation comparison ran at the tier where that ceiling binds, so the contact model whose advantage lies above 800 Hz was tested where it cannot act. The negative result stands as a caution: a distance that ranks candidates correctly can still be measuring the microphone geometry.

Working-paper status. The fitted numbers in Section 3 were measured through the defect identified in Section 4.1 and are reported as the evidence that prompted the audit, not as results. Section 5 lists what must be re-derived before any of them is believed.

1. WHAT IS BEING FITTED

The instrument is a real-time modal model of a two-headed tom: two circular membranes coupled through a lumped air cavity, with a Berger tension nonlinearity, a microphone model, and a three-band stochastic attack layer standing in for the unresolvably dense high-frequency region. Seventeen normalised parameters are exposed to the product — diameter, both tensions, damping and its frequency tilt, strike and pickup geometry, shell depth, cavity coupling, nonlinearity, asymmetry, and the attack layer’s level and corner frequency — and the fit searches exactly that bank, so that whatever it finds is reachable by a user.

The target is a single recorded tom hit of **unknown provenance**: a sample-library file with no record of the drum, head, tuning, mallet, microphone, room or gain chain, and no licence permitting redistribution. It is therefore a timbre-match target, not an acoustic validation reference, and no automated test depends on it. That distinction turns out to matter more than it was expected to.

2. METHOD

Reference and candidate are reduced to the same feature vector and scored by an eight-term distance, each term in its own perceptual unit so that it can be read against a tolerance rather than only against another run: partial frequency (cents), partial level and spectral envelope (dB), partial decay (log ratio of T_{60}), amplitude envelope (dB), pitch glide (cents), attack balance (dB), and the share of reference partial energy left unmatched. Weights are the reciprocal of each term’s “clearly wrong” threshold, so a just-audible error anywhere contributes about equally. Both signals are peak-normalised, so every term is gain-invariant — forced, since the reference is normalised and its loudness carries no information.

Partials are found by prominence-based peak picking on a 64k transform of the sustain window, capped at sixteen, and matched greedily by closeness in cents with a tolerance that widens with mode index, since real two-headed drums scatter about $\pm 20\%$ around the ideal Bessel series in both directions [1].

The search minimises this distance with the Mayfly Optimization Algorithm [2]: eight independent restarts of 150 iterations at population 16, 59 056 objective evaluations per run, deterministic given the seed. Two excitation models were compared. `ContactPrescribed` writes a half-sine of the measured contact duration into a force buffer, so the head cannot influence it. `ContactHertzian` integrates the stick as a free mass against a Hunt–Crossley contact spring [3], [4], calibrated against measured contact times [5], [6].

3. WHAT THE FITS REPORTED

Term	Prescr.	Hertz.	Hertz. 5 g	Gate
Partial frequency	20.6 ¢	19.6 ¢	27.2 ¢	≤ 25 ¢
Partial decay	0.179	0.493	0.188	≤ 0.25
Spectral envelope	12.3 dB	14.5 dB	11.9 dB	≤ 4 dB
Partial level	2.0 dB	2.0 dB	2.0 dB	—
Amplitude envelope	1.0 dB	2.8 dB	3.1 dB	—
Glide	18.0 ¢	0.1 ¢	0.0 ¢	—
Attack balance	0.0 dB	0.0 dB	1.1 dB	—
Total	5.901	7.450	6.548	—
Partials in 476–700 Hz	0	0	0	9 in ref.

Table 1: The three fits, each 8 restarts $\times$ 150 iterations. The prescribed excitation wins on total and on two of three gate terms; none comes within three times the spectral-envelope gate. The 5 g row is the Hertzian model with the mallet reduced from the shipped 15 g to the mass its own velocity law implies.

The ordering was confirmed by ear — prescribed closest, then the 5 g Hertzian — which is the observation that makes the rest of this paper necessary: **the distance ranks correctly and none of the candidates is close**. A metric can be monotone in quality and still be measuring the wrong thing.

Two structural observations came out of the same runs. The Hertzian fit leaves the attack layer’s corner frequency ATK.T at 3426 Hz against the prescribed fit’s 1261 Hz, confirming a prediction that a physically excited high band removes the search’s need to fake one with noise. And the 5 g run achieves the **best** spectral envelope of any fit while finding the **fewest** partials — three — by parking ATK.T at 626 Hz, inside the disputed band, with the attack level raised 3.7-fold. Broadband noise satisfies a band-energy metric and cannot make resolvable partials. This is the argument for a per-term gate: on total alone that run would have looked like the best match to the recording.

4. THREE DEFECTS, FOUND BY AUDITING THE MEASUREMENT

4.1. The reference downmix is a comb filter

The reference is a two-channel file. Cross-correlating the channels shows they are the same signal: correlation **0.942** at a lag of **69 samples** (1.56 ms at 44.1 kHz), against 0.361 at zero lag. The harness reduced multi-channel input to mono by averaging, and averaging a signal with a delayed copy of itself is a comb filter [7],

$$H(f) = 2|\cos(\pi f\tau)|, \quad (1)$$

with nulls at $(2k+1)/2\tau$ and maxima at k/τ . For $\tau = 1.565$ ms those fall at 320, 959 and 1598 Hz, and at 639 and 1278 Hz respectively. Figure 1 measures the downmix against a single channel and finds exactly that envelope.

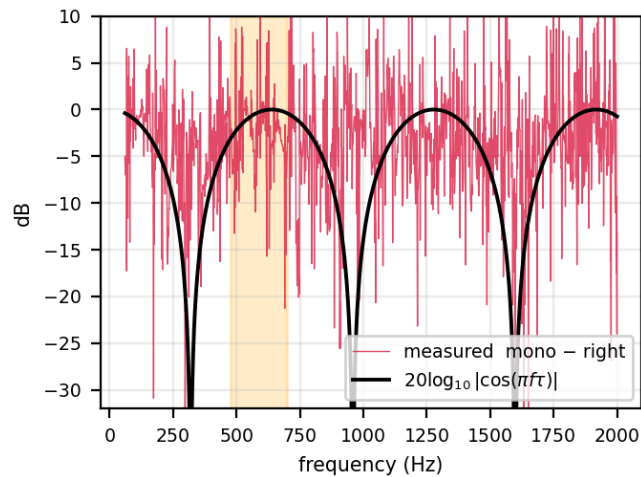


Figure 1: The mono downmix relative to the right channel alone, against Equation 1 with no fitted parameter. Shaded: the 476–700 Hz band.

The consequence is not subtle. The comb’s first null at 320 Hz hollows out the 200–476 Hz region, and its first maximum at 639 Hz sits in the middle of the 476–700 Hz band. With the partial detector capped at sixteen peaks, that shifts which peaks survive:

Reference analysed as	Partials	In 476–700 Hz
mono, averaged — <i>every fit above</i>	16	9
right channel alone	7	2
mono, aligned then averaged	5	2

Table 2: The same recording, three reductions. Aligning before summing brings the mono view into agreement with the single channels; the naive sum does not agree with either.

The dense cluster of nine partials between 476 and 700 Hz — documented as the recording’s essential character, and treated as the model’s outstanding deficiency because no mode count or excitation could reproduce it — is substantially an artefact of the sum. De-combed, the reference has five detected partials, two of them in that band. The model produces five.

The same defect distorted the decay measurements. Figure 2 shows the combed reference’s ring times against the $T_{60} \propto 1/f$ that constant Q requires of a membrane [8]: in the disputed band the median is 1.13 s where constant Q predicts 0.31 s, a factor of 3.6. Read at the time as evidence of a reverberant room, it is at least partly the comb’s nulls beating against the partials whose decay is being fitted — the aligned reduction gives 0.16, 0.19, 0.60 and 0.61 s, which are membrane-like.

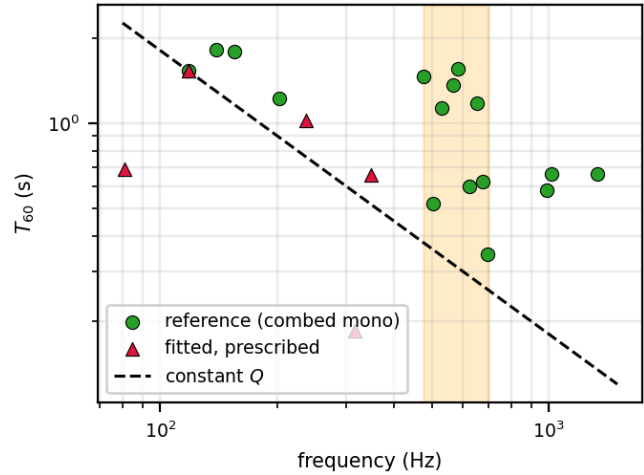


Figure 2: Ring time against frequency for the combed reference and the prescribed fit, with constant Q anchored on the reference fundamental.

The clue was present and read backwards. The provenance notes record “channel correlation 0.36 — there is a room on it”. The correct reading of a zero-lag correlation that low between two views of one close-miked hit is that the channels are time-offset, and must not be summed.

4.2. The modal bandwidth collapses when the drum is tuned down

Mode selection retains a fixed **count** — 48, 96 or 160 by quality tier — and membrane mode frequencies scale as $\sqrt{T}$. The fit lowers batter tension from the shipped 1250 N/m to 331 N/m, because tension is the parameter that lengthens the fundamental’s ring at unchanged damping, and that halves the ceiling:

Tier	Modes	Top mode	In 476–700 Hz
Draft — <i>used by the search</i>	48	467 Hz	0
Standard	96	664 Hz	45
High	160	852 Hz	58

Table 3: The fitted bank’s modal ceiling by quality tier.

The fitted drum has no modes at all above 467 Hz (Figure 3), and the quality parameter is pinned during the search, so the optimiser cannot buy them back. This is a modelling defect independent of the fit: a real membrane has modes at every frequency, and truncating by count makes the rendered bandwidth a function of the tuning, so the instrument loses its top octave when it is tuned down.

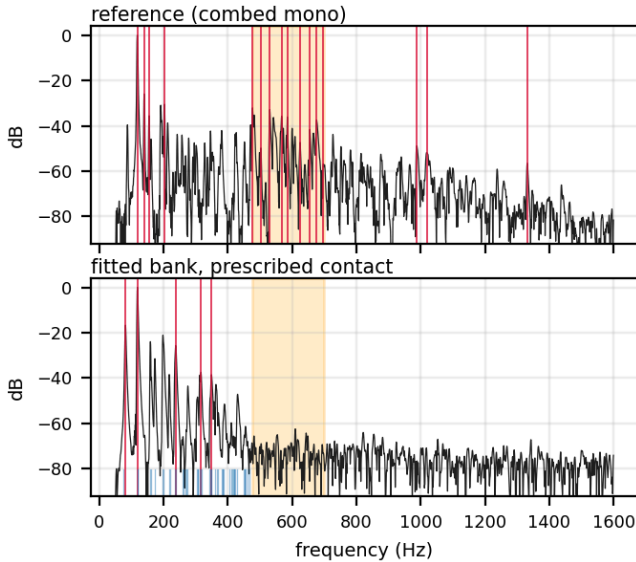


Figure 3: Sustain spectra. Red: detected partials. Blue ticks along the axis: the model’s retained modes, which stop at 467 Hz. Above that the fitted bank has only the attack layer’s noise floor.

4.3. The excitation comparison was confounded

The Hertzian contact’s measured advantage is +11.8 dB at 800 Hz, +15.1 at 1.5 kHz and +22.9 at 2.5 kHz, and only 0–4 dB below 700 Hz. Both fits ran at the Draft tier, where the modal bank stops at 467 Hz — so the improved excitation had almost nothing modal to drive, and its whole advantage landed in a region the model does not populate. Re-measuring the fitted banks at Standard, the Hertzian configuration is the only one that has ever produced partials inside the disputed band (three to four, reaching 1000 Hz), while the prescribed configuration gains nothing from the extra modes, as its spectral zeros predict.

The conclusion that the Hertzian contact does not earn its calibration pass is therefore not supported. It was tested where it cannot act.

5. CONSEQUENCES

Everything measured through the naive downmix must be re-derived: the fitted banks, the committed fixture and preset, and the claim that the model cannot populate 476–700 Hz. That claim was the highest-priority open item on this model’s roadmap, and the evidence for it is now largely attributable to the measurement.

Three changes follow. First, the mono reduction measures the inter-channel lag and aligns before averaging, guarded so that it fires only when the pair is genuinely one signal twice — correlation at least 0.5 and clearly better than summing — and never on short or uncorrelated input; the measured lag and correlation are reported on the decoded reference so the condition cannot hide again. Second, mode selection should be bounded by frequency rather than by count, so that the ceiling does not move with the tuning. Third, the excitation comparison must be repeated at a tier where the modal bank reaches the band the contact model acts on.

The wider caution is the one this paper is named for. The fit was well-behaved throughout: it converged, its restarts spread sensibly, its ranking was stable across contact models and mallet masses, and a listening test agreed with its ordering. None of that constrains whether the target is the instrument. A distance is only as good as the reduction that produced it, and the reduction here imposed a ± 6 dB spectral shape — a null and a peak, both fixed by microphone spacing — on every number in Table 1.

6. SCOPE AND REPRODUCIBILITY

The model, the feature extraction, the distance and the search are in `internal/physical`, `internal/physical/match` and

`cmd/fit-physical`. Runs are deterministic given the seed and are excluded from continuous integration, since they take minutes and require a recording the repository does not contain. Figures here are generated from the reference and from the committed fit reports; the recording itself is not redistributable and is not part of the repository. The channel-alignment behaviour is pinned by regression tests in both directions — that a delayed pair is aligned, and that an uncorrelated pair is left alone.

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